

SML

Assignment - 2

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1 Objective

The objective of this assignment is to implement dimensionality reduction techniques and classification methods on the MNIST dataset. Principal Component Analysis (PCA) and Fisher Discriminant Analysis (FDA) are used to reduce the dimensionality of the data. The reduced feature representations are then used for classification using Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA). The goal is to evaluate how dimensionality reduction affects classification performance.

2 Dataset Preparation

The MNIST dataset consists of grayscale images of handwritten digits. Each image has a size of 28×28 pixels. Each image is converted into a feature vector of dimension 784 by flattening the image. The preprocessing steps are:

1. Load the MNIST training and test datasets.
2. Filter the dataset to keep only digit classes 0, 1, and 2.
3. Normalize pixel values to the range $[0, 1]$ by dividing by 255.
4. Convert each image into a 784-dimensional vector.
5. Randomly select 100 samples from each class for training and testing.

Final dataset statistics:

- Training samples: 300
- Test samples: 300
- Feature dimension: 784

3 Maximum Likelihood Estimates

For classification, the mean vector and covariance matrix are estimated from the data.

3.1 Mean Vector

The mean vector is computed as

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

where N is the number of samples and x_i is the feature vector.

3.2 Covariance Matrix

The covariance matrix is computed as

$$\Sigma = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)(x_i - \mu)^T$$

The covariance matrix captures the spread and relationships between features.

4 Principal Component Analysis (PCA)

PCA is used to reduce dimensionality while preserving the variance of the data.

4.1 Mean Centering

The mean vector is computed and subtracted from each sample:

$$X_c = X - \mu$$

4.2 Covariance Matrix

The covariance matrix is computed as

$$S = \frac{1}{N-1} X_c X_c^T$$

4.3 Eigen Decomposition

The eigenvalues and eigenvectors of the covariance matrix are computed:

$$S u_i = \lambda_i u_i$$

Eigenvectors corresponding to the largest eigenvalues capture the most variance.

4.4 Projection

The data is projected onto the principal components:

$$Y = U_p^T X_c$$

where U_p contains the selected eigenvectors.

Three PCA settings were evaluated:

- Retaining 75% of the total variance
- Retaining 90% of the total variance
- Using only the first 2 principal components

5 Fisher Discriminant Analysis (FDA)

FDA is a supervised dimensionality reduction method that maximizes class separability.

5.1 Between-Class Scatter Matrix

$$S_B = \sum_c N_c (\mu_c - \mu)(\mu_c - \mu)^T$$

where

- μ_c is the mean of class c
- μ is the overall mean
- N_c is the number of samples in class c

5.2 Within-Class Scatter Matrix

$$S_W = \sum_c \sum_{x_i \in C_c} (x_i - \mu_c)(x_i - \mu_c)^T$$

5.3 Optimal Projection

FDA maximizes the ratio

$$\frac{\det(W^T S_B W)}{\det(W^T S_W W)}$$

This leads to the generalized eigenvalue problem:

$$S_W^{-1} S_B w = \lambda w$$

The eigenvectors corresponding to the largest eigenvalues form the projection matrix W .
The projected data is

$$Y = XW$$

6 Linear Discriminant Analysis (LDA)

LDA assumes that all classes share a common covariance matrix.

The discriminant function is

$$g(x) = \mu^T \Sigma^{-1} x - \frac{1}{2} \mu^T \Sigma^{-1} \mu$$

The predicted class is the one with the maximum discriminant score.

7 Quadratic Discriminant Analysis (QDA)

QDA allows each class to have its own covariance matrix.

The discriminant function is

$$g(x) = -\frac{1}{2} \log |\Sigma| - \frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)$$

This results in quadratic decision boundaries between classes.

8 PCA Reconstruction

PCA can also be used to reconstruct the original data.

First the projected representation is computed:

$$Y = U_p^T (x - \mu)$$

The reconstructed sample is

$$x_{reconstructed} = U_p Y + \mu$$

Reconstruction quality is measured using Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum (x - x_{reconstructed})^2$$

9 Results

9.1 PCA + LDA

PCA Setting	Train Accuracy	Test Accuracy
75% Variance Retained	0.97	0.97
90% Variance Retained	0.98	0.97
2 Principal Components	0.93	0.9167

Table 1: Classification accuracy using PCA features with LDA

9.2 PCA + QDA

PCA Setting	Train Accuracy	Test Accuracy
75% Variance Retained	0.9867	0.9567
90% Variance Retained	0.9933	0.9067
2 Principal Components	0.94	0.93

Table 2: Classification accuracy using PCA features with QDA

9.3 PCA Reconstruction Error

Sample	MSE
1	0.0841
2	0.0707
3	0.0487
4	0.0624
5	0.0428
Average	0.0618

Table 3: Mean Squared Error between original and reconstructed samples

9.4 FDA Results

Method	Train Accuracy	Test Accuracy
FDA + LDA	1.00	0.88
FDA + QDA	1.00	0.88

Table 4: Classification accuracy using FDA projection

10 Observations

PCA effectively reduces dimensionality while preserving variance in the data. However, PCA does not utilize class label information during dimensionality reduction.

FDA incorporates class labels and produces projections that maximize class separability.

LDA performs well when the assumption of shared covariance across classes holds. QDA is more flexible but requires estimating separate covariance matrices for each class.

11 Conclusion

In this assignment, PCA and FDA were implemented for dimensionality reduction on the MNIST dataset containing digits 0, 1, and 2. The reduced features were then used with LDA and QDA classifiers.

The experiments demonstrate that PCA efficiently compresses high-dimensional data, while

FDA produces projections that are better suited for classification tasks due to its supervised nature. Combining dimensionality reduction with probabilistic classifiers provides an effective approach for handwritten digit classification.

12 Visualization

Two-dimensional visualizations were generated using PCA and FDA projections. PCA projections display the data along directions of maximum variance. FDA projections show better class separation because FDA explicitly maximizes the distance between class means while minimizing within-class variance.

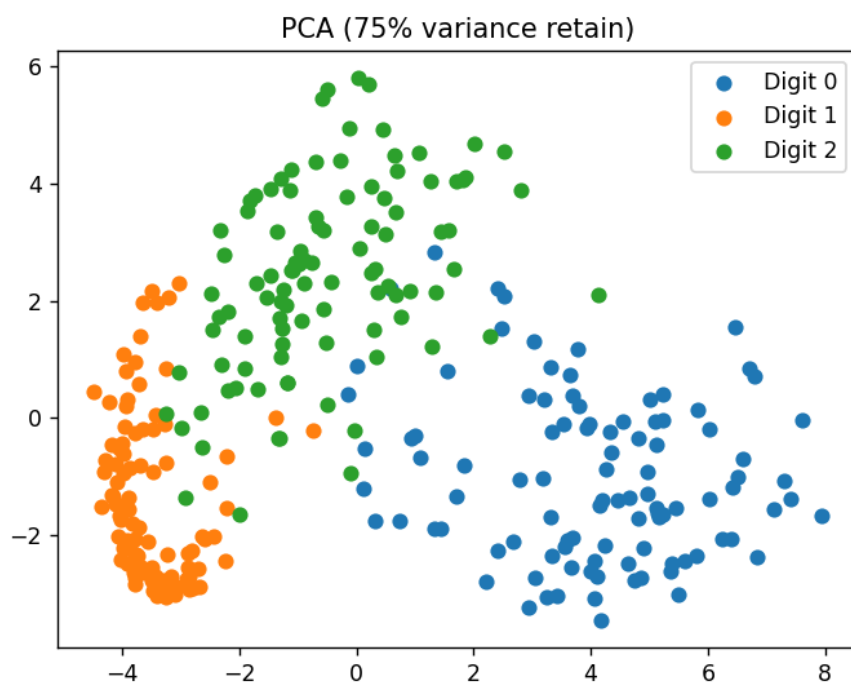


Figure 1: PCA projection of the dataset retaining 75% variance. Each point represents an image projected onto the first two principal components. The colors correspond to digit classes (0, 1, 2).

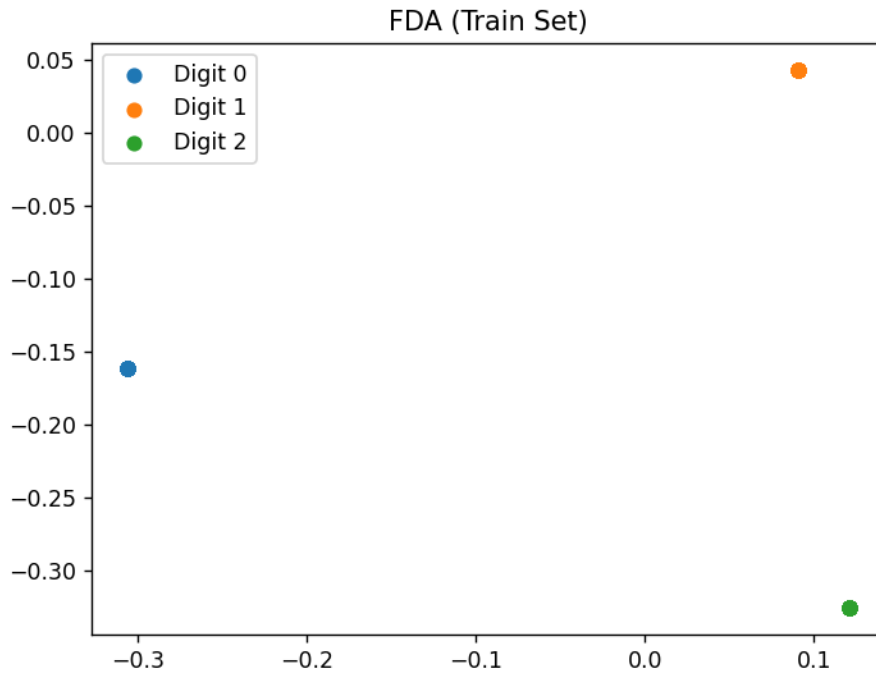


Figure 2: FDA projection of the training data. FDA maximizes class separability by projecting the data onto directions that maximize between-class scatter and minimize within-class scatter.

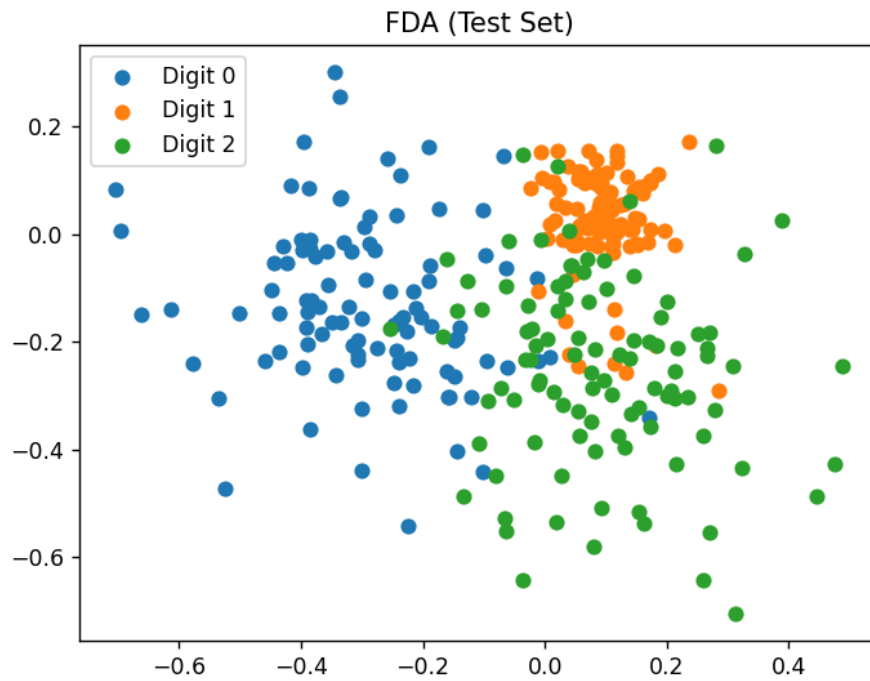


Figure 3: FDA projection of the test dataset using the projection matrix learned from the training data.